Indoor Pollution

Pupil sheet

Name: _____ **Date:** _____

Safety: Only the teacher opens the window; never climb on the sill, a radiator or a chair.

1. What I see on the screen

The screen shows three cards: **CO_2** (in ppm), **Temperature** (°C) and **Humidity** (%). The sensor gives a new value **once every five seconds**, so the line climbs in steps.

Three rules. I stay **at least a metre** from the box — otherwise I measure my own breathing, not the room. I record the **time** alongside the number; without it I cannot compute any rate. And I record **how many of us there are** and whether the window is open.

Baseline (aired, empty room): _____ ppm **Outdoors:** _____ ppm **We are:** _____

2. What I do on my own

First I predict. I do not change my prediction after I measure.
In 15 minutes, with the whole class inside, the CO_2 will rise by about _____ ppm.
After we open the window, it will come back in about _____ minutes.

because _____

_____

The rise — window closed, everybody in the room.

Minute	CO_2 (ppm)	T (°C)	Humidity (%)
0			
5			
10			
15			

The recovery — window open. ☐ and the door

Minute	CO_2 (ppm)	T (°C)	Humidity (%)
0			
1			
2			
3			
5			

I compute the rates. Worked example: $1150 - 700 = 450$ ppm in 15 min, so $450 : 15 = 30$ ppm/min.

The rise: _____ − _____ = _____ ppm in _____ min, so _____ ppm/min
The recovery: _____ − _____ = _____ ppm in _____ min, so _____ ppm/min
The app: the rise _____ ± ___, the recovery _____ ± ___ ppm/min. A match? ☐ yes ☐ no

3. The conclusion I write

I predicted that, and the data show that _____

_____

The rise eased off after a while because _____

_____

Our ventilation recommendation. How many times an hour, and for how long? Use both rates.

_____

The limit of our measurement: what does this sensor *not* see?

_____

Exit ticket

How many times an hour should our room be aired, and how do you know?

_____